



KAPWING



Pinky and the Brain

Ed Wood
capybara



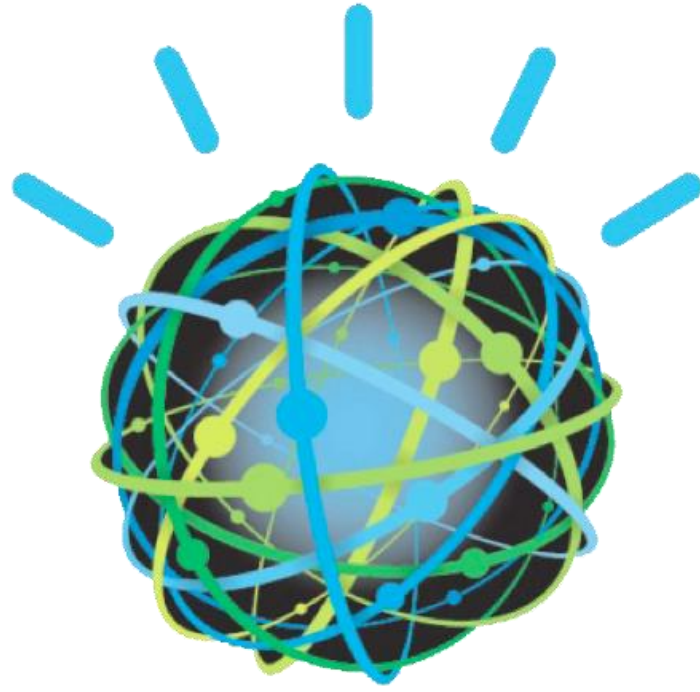
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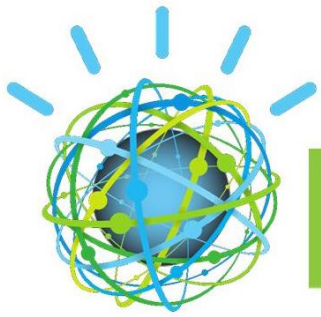
\$15,000

\$



IBM[®] WATSON





IBM Watson

IBM

WATSON

IBM
Araştırma Projesi
(2006)



AR-GE

Jeopardy!
(Şubat, 2011)



Uygulama

Sağlık
Uygulamaları
(Ağustos, 2011)



Ticarileşme



Finans
Uygulamaları
(Mart, 2012)

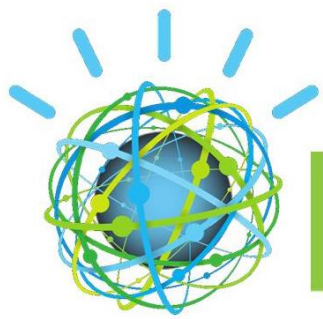


Büyüme

Endüstriyel
Çözümler
(2012, -)



Farklı Platformlar



IBM Watson





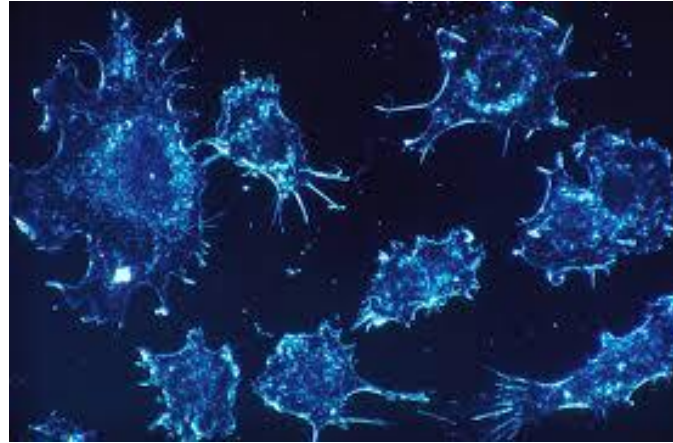
Sağlıkta yapay zeka





Sağlıkta yapay zeka

- Kanser

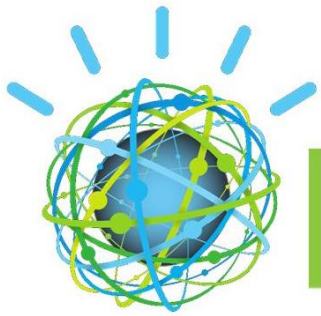




Sağlıkta yapay zeka

- Kanser
- Diyabet



The IBM Watson logo, featuring the word "IBM" in its characteristic eight-colored stripes and the word "Watson" in a blue serif font.

Sağlıkta yapay zeka

- Kanser
- Diyabet
- Spor & Sağlık





Finans sektöründe yapay zeka





Finans sektöründe yapay zeka

1. Yasal zorunluluklar
2. Finansal suçlar
3. Risk modellemesi





Ve sektöre yönelik diğer çözümler...





Ve sektöre yönelik diğer çözümler...

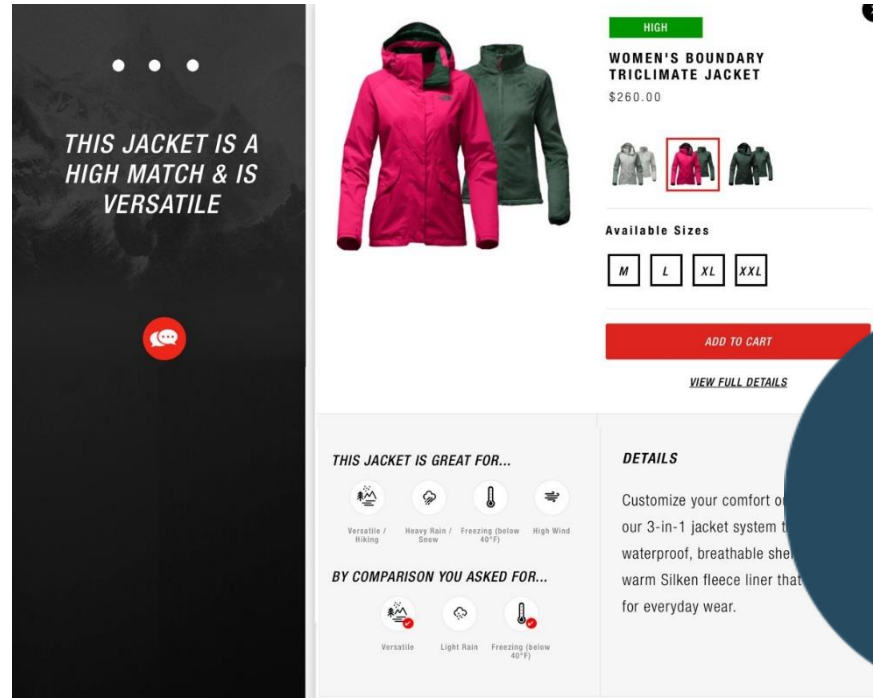
- Siber suçlarla mücadele





Ve sektöre yönelik diğer çözümler...

- Siber suçlarla mücadele
- Kişisel alışveriş





Ve sektöre yönelik diğer çözümler...

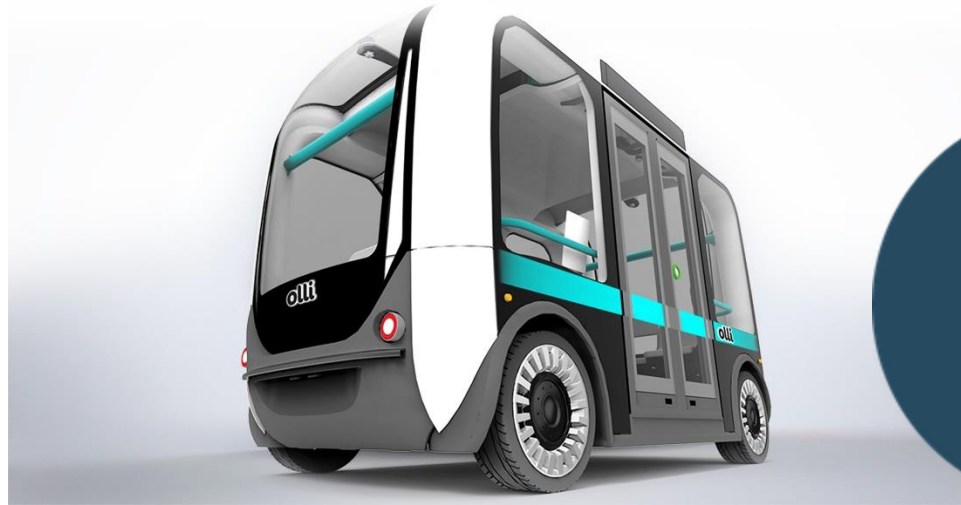
- Siber suçlarla mücadele
- Kişisel alışveriş
- Otonom araçlar





Ve sektöre yönelik diğer çözümler...

- Siber suçlarla mücadele
- Kişisel alışveriş
- Otonom araçlar

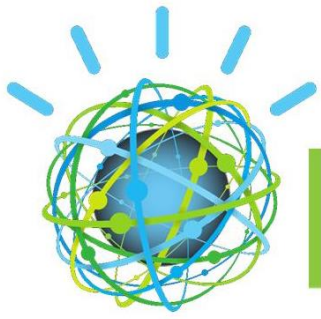




Ve sektöre yönelik diğer çözümler...

- Siber suçlarla mücadele
- Kişisel alışveriş
- Otonom araçlar
- Akşam yemeği 😊

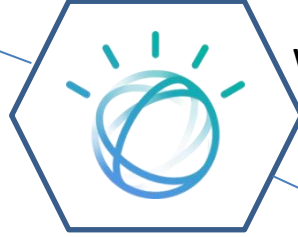




IBM Watson



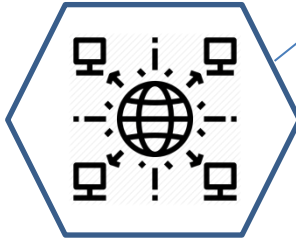
Kullanıcıdan mesajı alır



Watson'la isteği çözümler / anlar



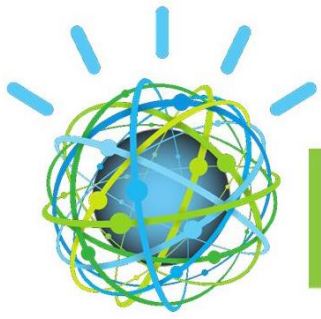
Cevabı üretir



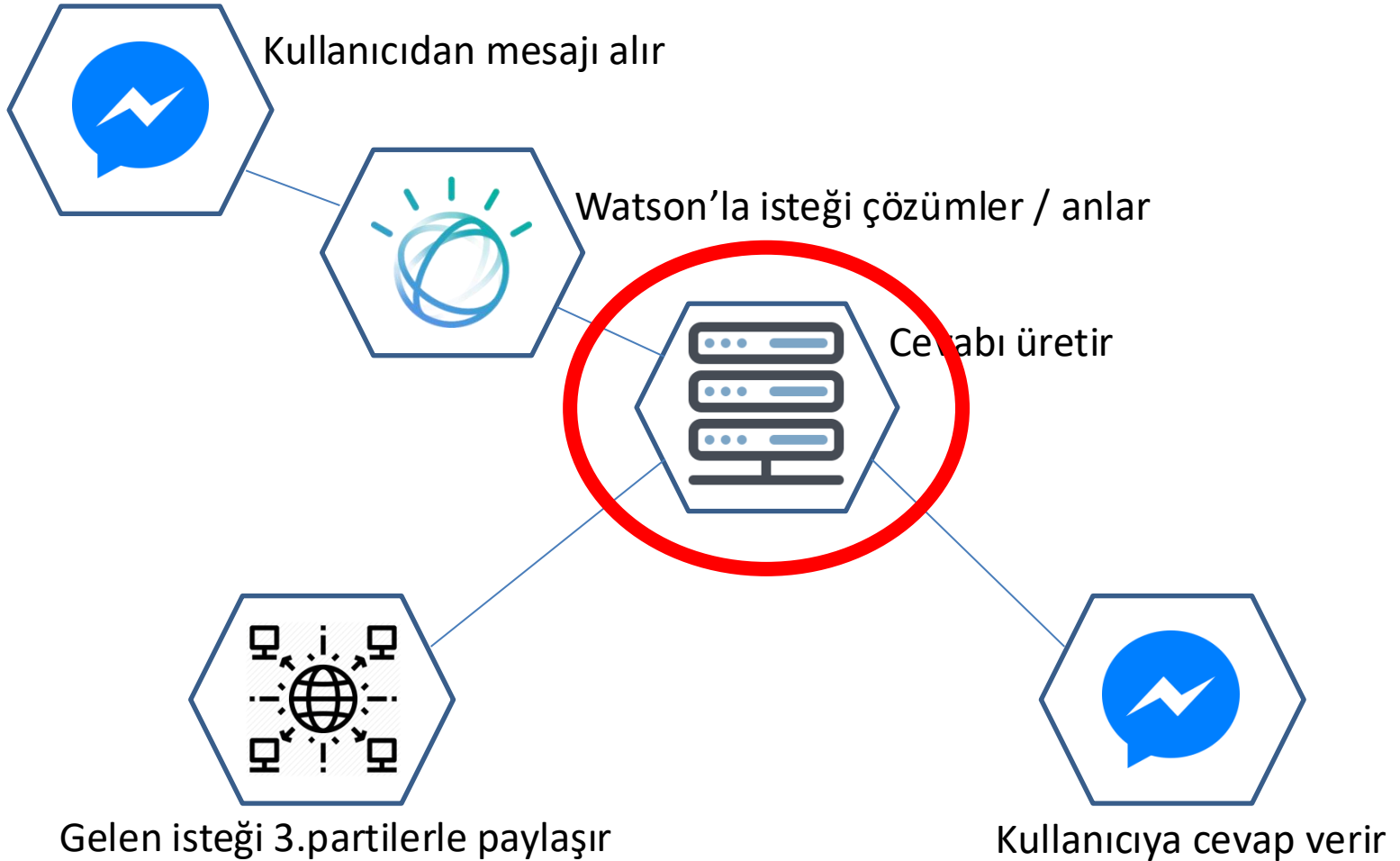
Gelen isteği 3.partilerle paylaşır

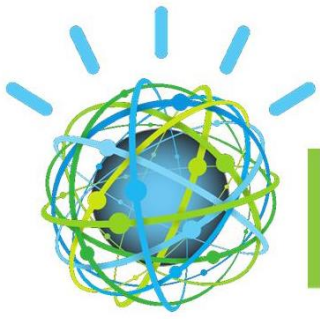


Kullanıcıya cevap verir

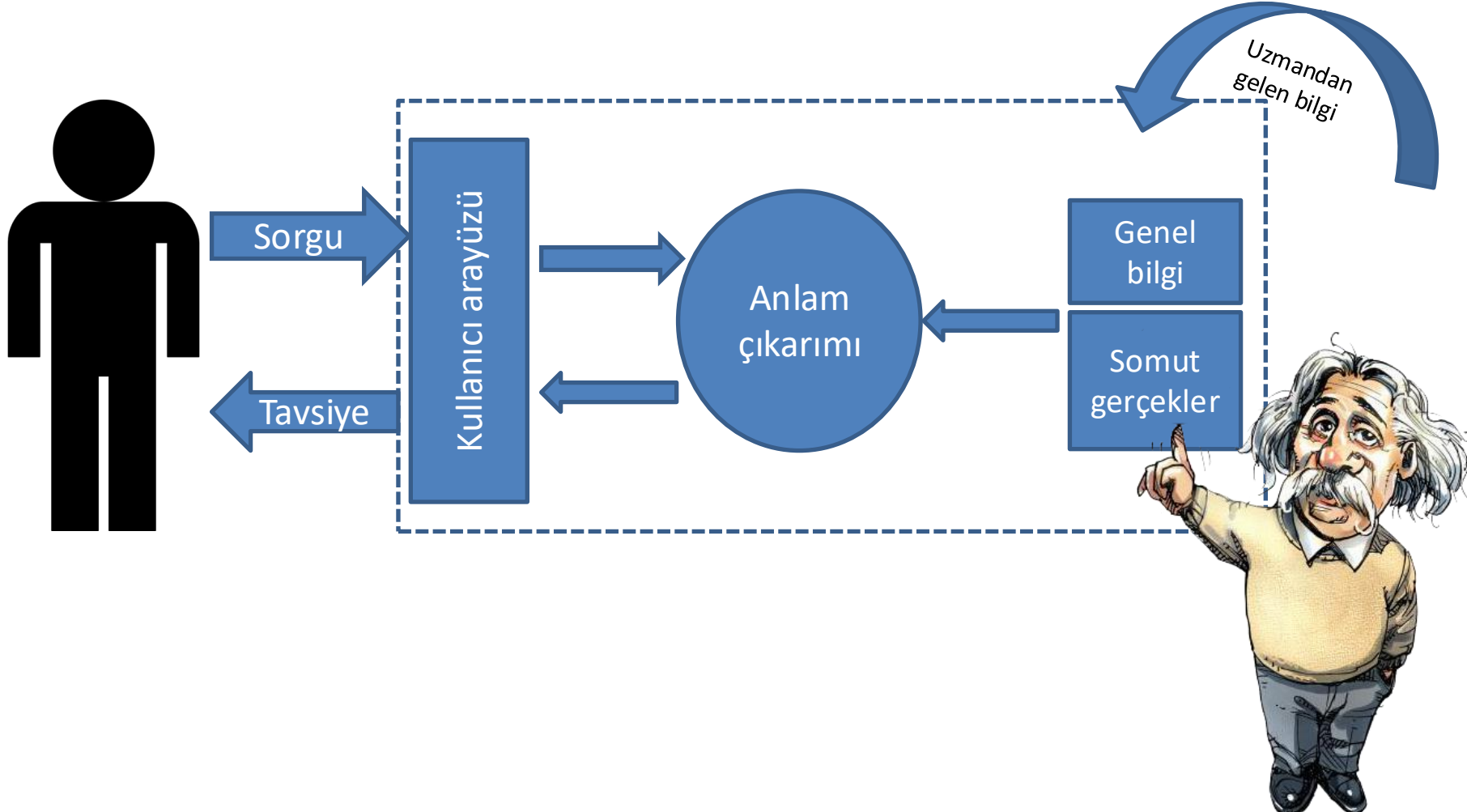


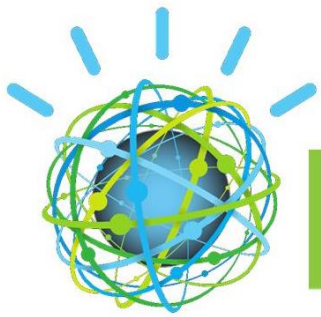
IBM Watson





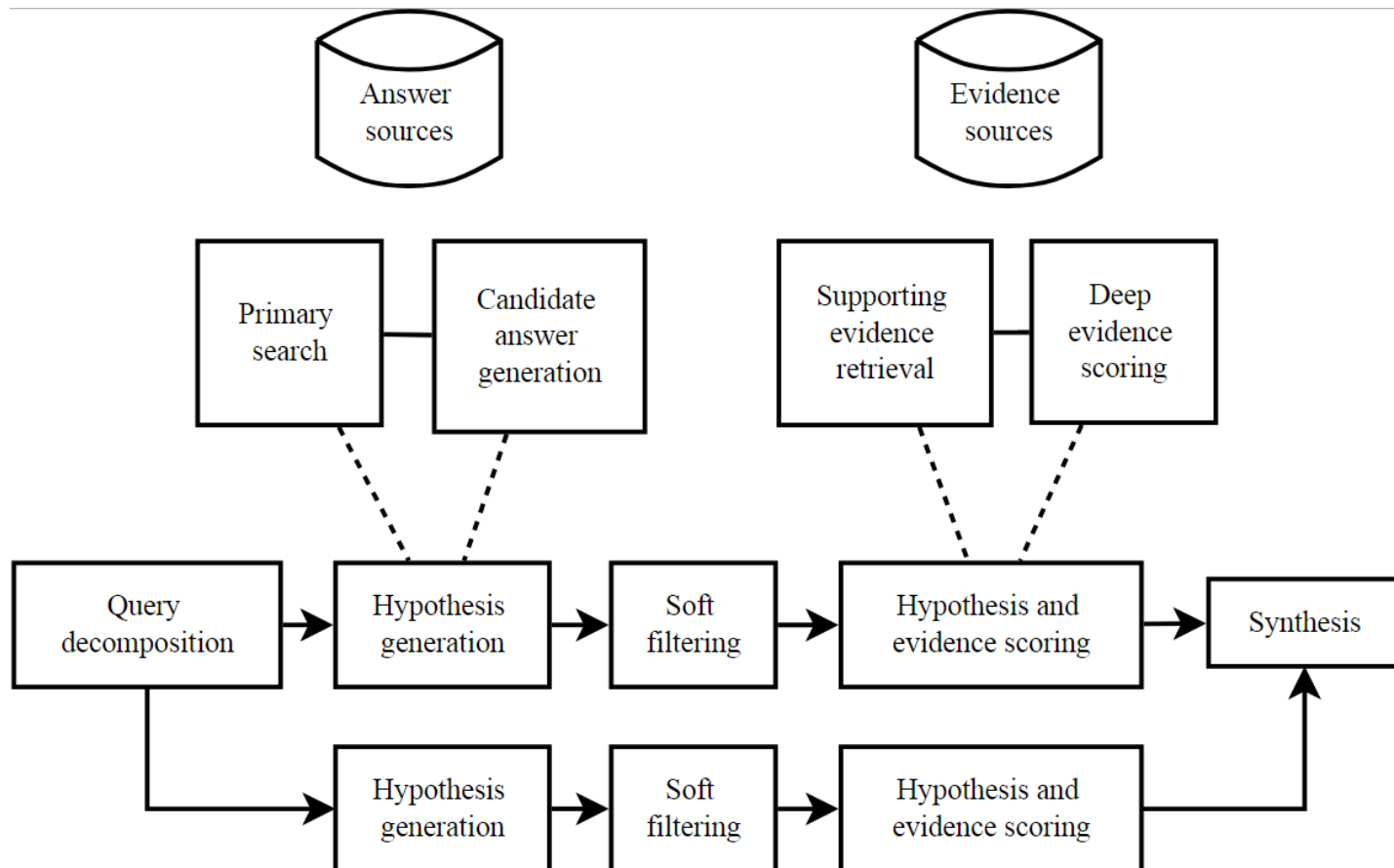
IBM Watson





IBM Watson

Anlam
çıkarmı





İnsanları değiştirmek ya da alt etmek için değil,
doğru kararları vermelerinde onlara yardım
etmek için geliştirildi.



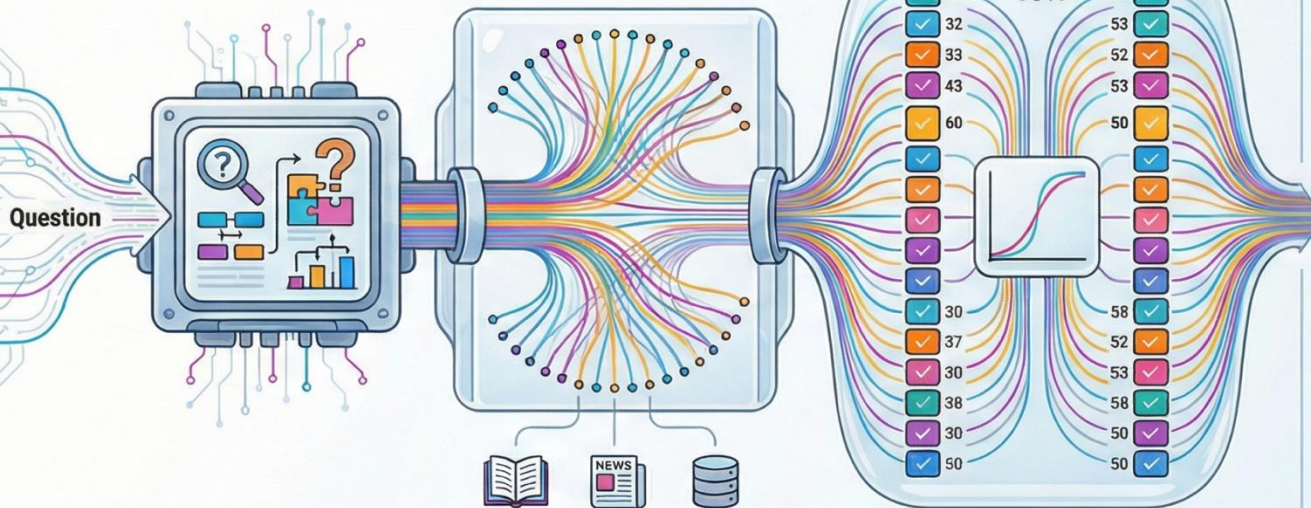
KAPWING

“Coping With Humans”

geçen haftaki tatsızlıktan sonra

Building Watson: The Architecture of a Jeopardy! Champion

DeepQA Processing Pipeline



Question Analysis & Decomposition

The system identifies the Lexical Answer Type (LAT) and decomposes complex clues into sub-clues.

Massively Parallel Hypothesis Generation

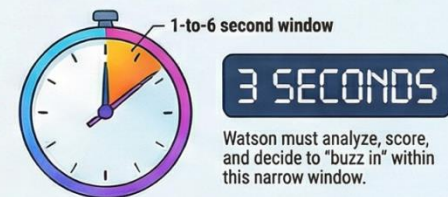
Watson generates hundreds of candidate answers from diverse sources like encyclopedias, newswire, and ontologies.

Ensemble Evidence Scoring & Ranking

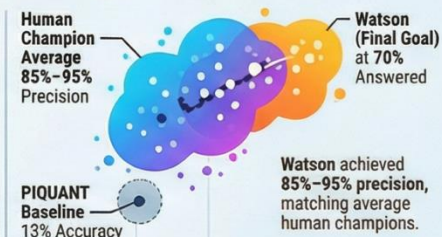
Over 50 scoring components evaluate candidates, using logistic regression to produce a final confidence score.

Performance Benchmarks vs. Humans

3-Second Response Time



Matching the "Winners Cloud"



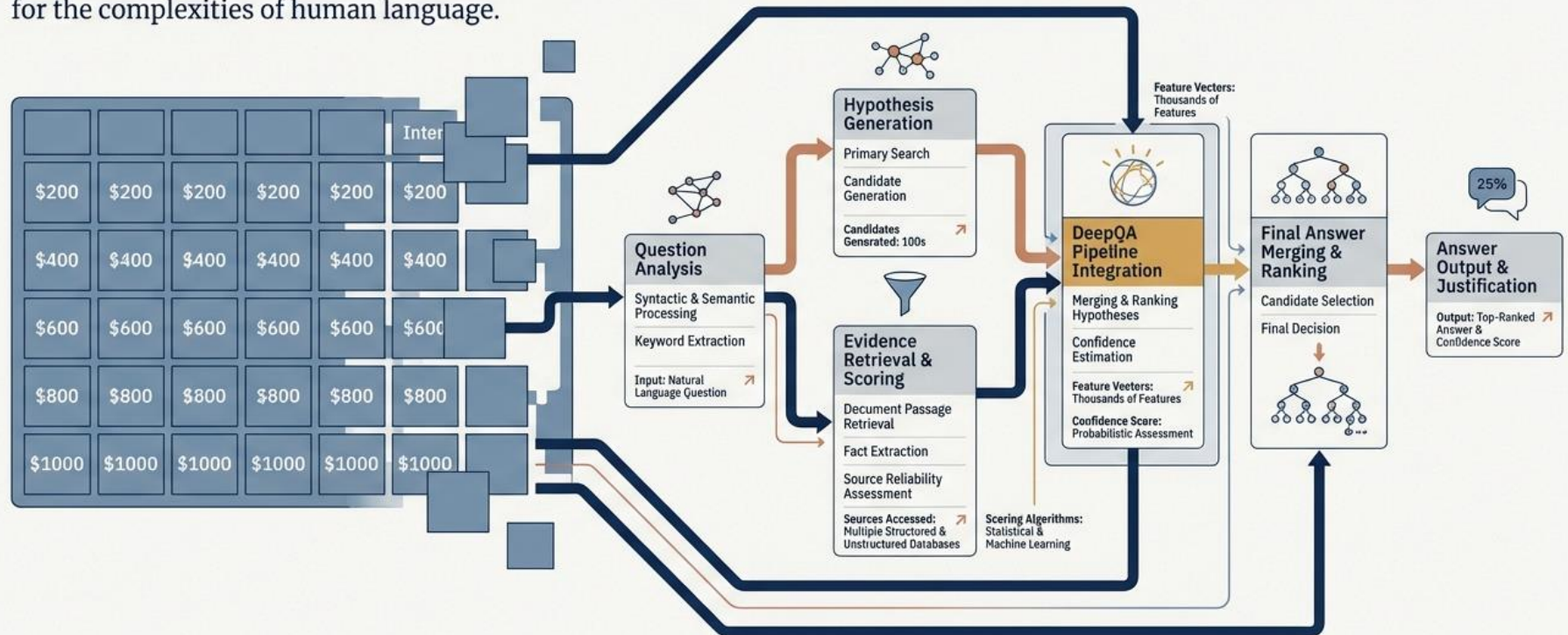
The Ken Jennings Standard



Building Watson: Inside the DeepQA Project

Infographic slide in Infogage in the DeepQA Project

The architecture that rivaled Deep Blue, built for the complexities of human language.



Keyword Search is Failing the Enterprise

In the enterprise, popularity is not as important an indicator of relevance, and recall can be as critical as precision.

Status Quo

keyword 1 + keyword 2



Retrieving popular documents based on 2-3 keywords.

The Future

Natural language question?



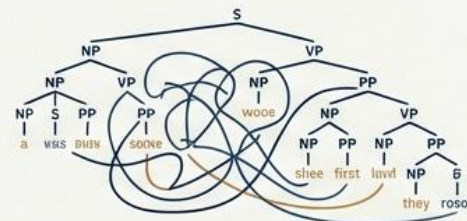
Synthesizing precise answers with justified confidence.

The Ultimate Test of Natural Language



Broad Domain

Covers everything from history and science to puns and wordplay.



Complex Language

Ambiguous, noisy, and layered phrasing.



High Stakes

Financial penalties for wrong answers require highly accurate confidence estimation.



Extreme Speed

Roughly 3 seconds to compute confidence and hit the buzzer.

Category: General Science

Clue: When hit by electrons, a phosphor gives off electromagnetic energy in this form.

Answer: Light (or Photons)

Category: Lincoln Blogs

Clue: Secretary Chase just submitted this to me for the third time; guess what, pal. This time I'm accepting it.

Answer: his resignation

Category: Head North

Clue: They're the two states you could be reentering if you're crossing Florida's northern border.

Answer: Georgia and Alabama

Category: "Rap" Sheet

Clue: This archaic term for a mischievous or annoying child can also mean a rogue or scamp.

Subclue 1: This archaic term for a mischievous or annoying child.

Subclue 2: This term can also mean a rogue or scamp.

Answer: Rascal

Category: Before and After Goes to the Movies

Clue: Film of a typical day in the life of the Beatles, which includes running from bloodthirsty zombie fans in a Romero classic.

Subclue 2: Film of a typical day in the life of the Beatles.

Answer 1: (*A Hard Day's Night*)

Subclue 2: Running from bloodthirsty zombie fans in a Romero classic.

Answer 2: (*Night of the Living Dead*)

Answer: *A Hard Day's Night of the Living Dead*

Category: Rhyme Time

Clue: It's where Pele stores his ball.

Subclue 1: Pele ball (soccer)

Subclue 2: where store (cabinet, drawer, locker, and so on)

Answer: soccer locker

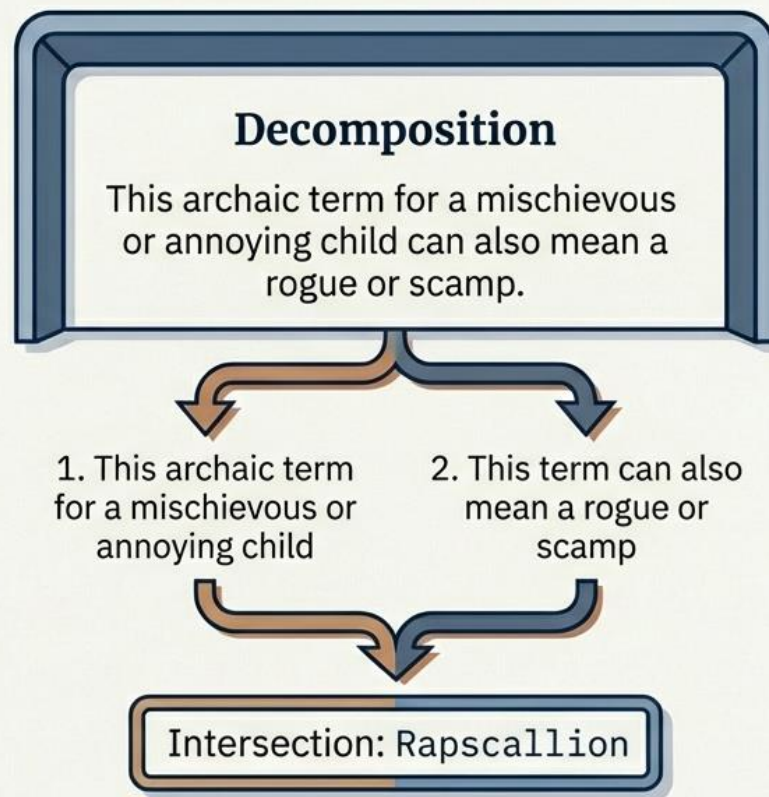
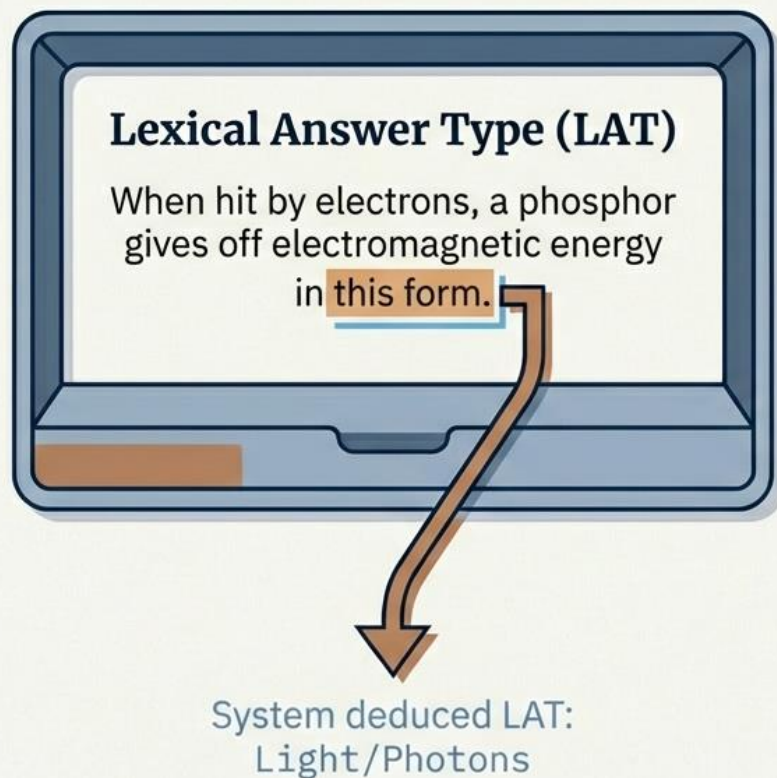
Category: Picture This

(Contestants are shown a picture of a B-52 bomber)

Clue: Alphanumeric name of the fearsome machine seen here.

Answer: B-52

The Anatomy of a Clue



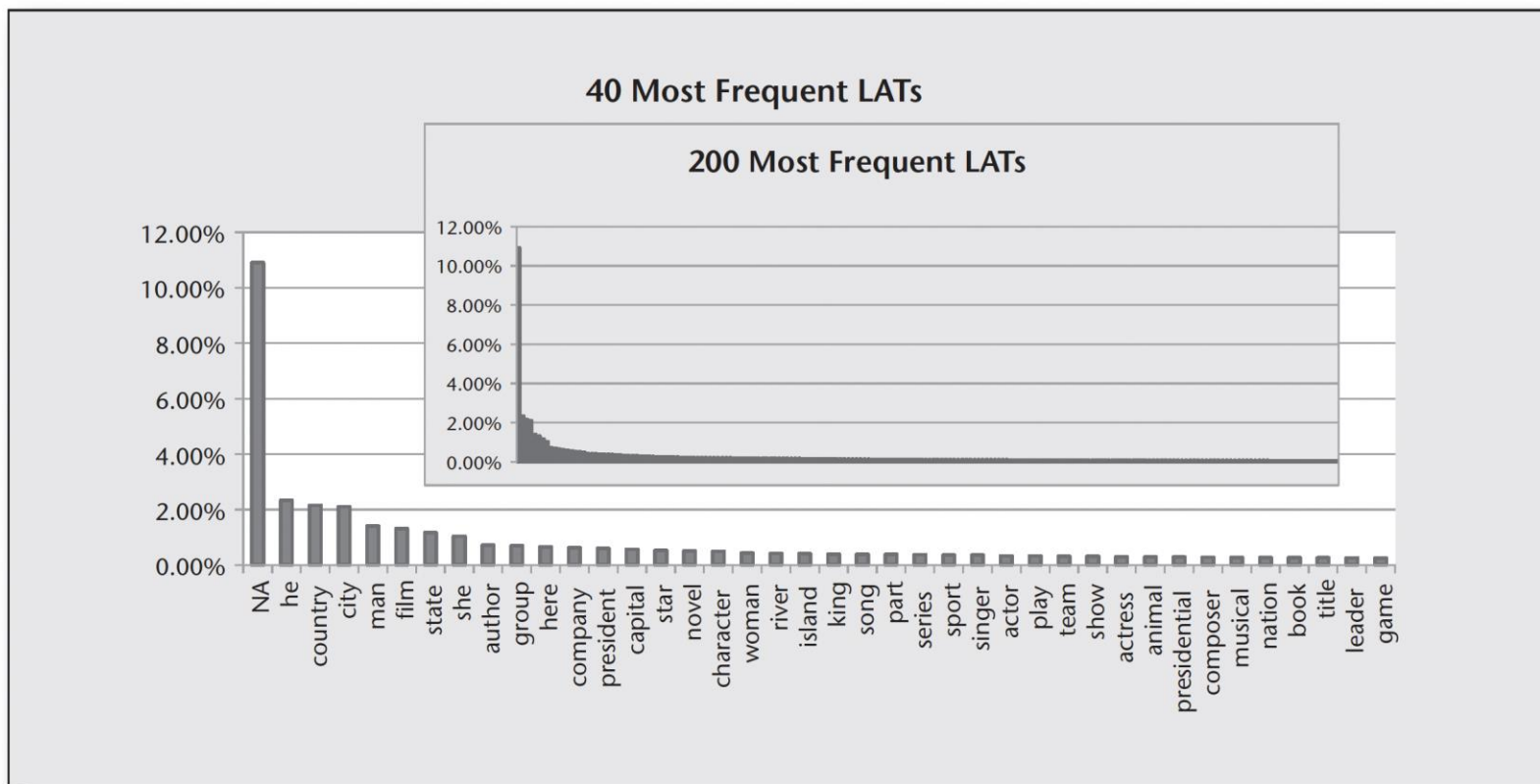


Figure 1. Lexical Answer Type Frequency.

*“As a measure of the Jeopardy Challenge’s breadth of domain, we analyzed a random sample of 20,000 questions extracting the **lexical answer type** (LAT) when present. We define a LAT to be a word in the clue that indicates the type of the answer, independent of assigning semantics to that word.”*

Category: Oooh....Chess

Clue: Invented in the 1500s to speed up the game, this maneuver involves two pieces of the same color.

Answer: Castling

LAT: *maneuver*

Benchmarks

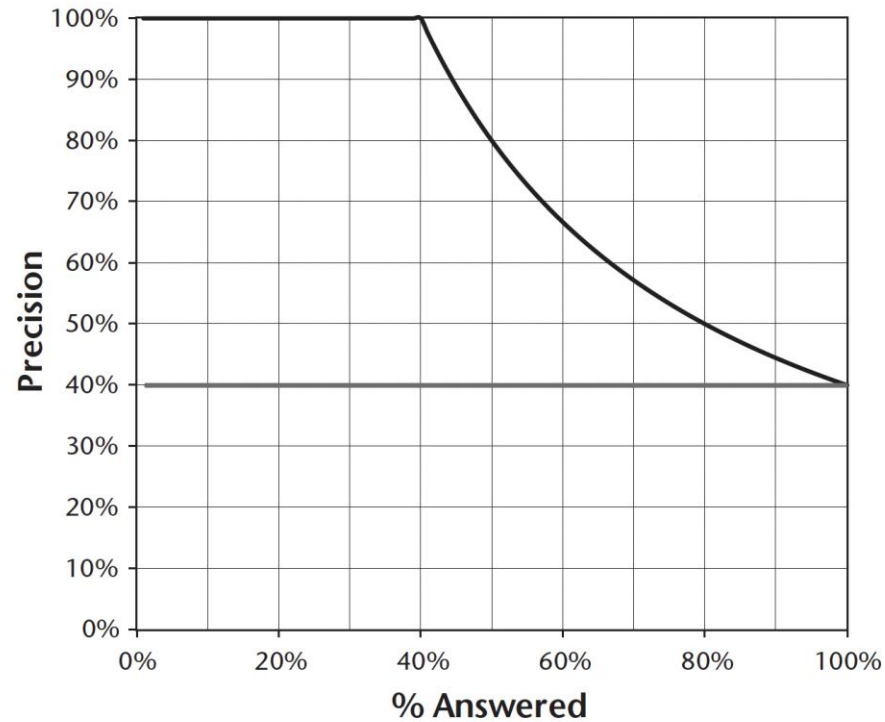
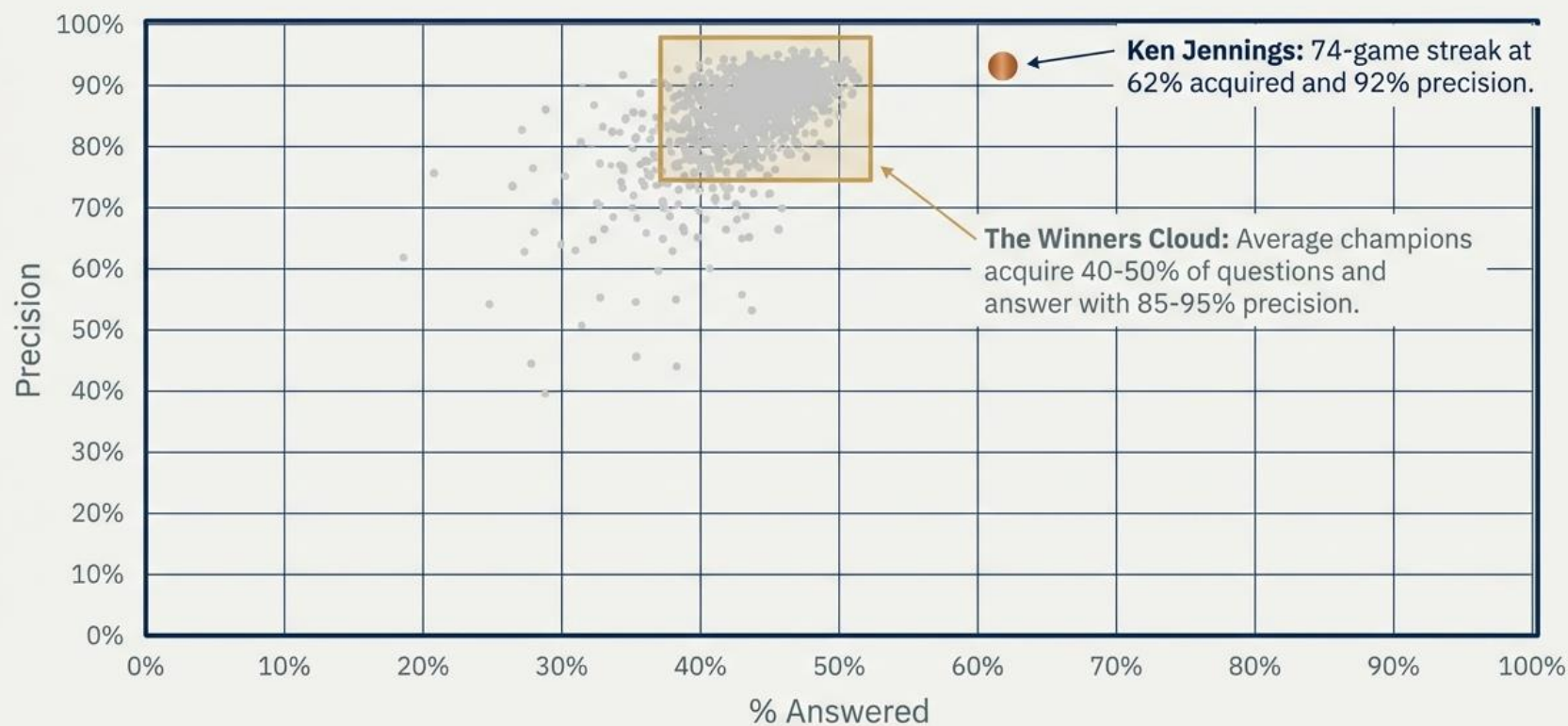


Figure 2. Precision Versus Percentage Attempted.

Perfect confidence estimation (upper line) and no confidence estimation (lower line).

Human Champions Set an Imposing Benchmark



Benchmarks

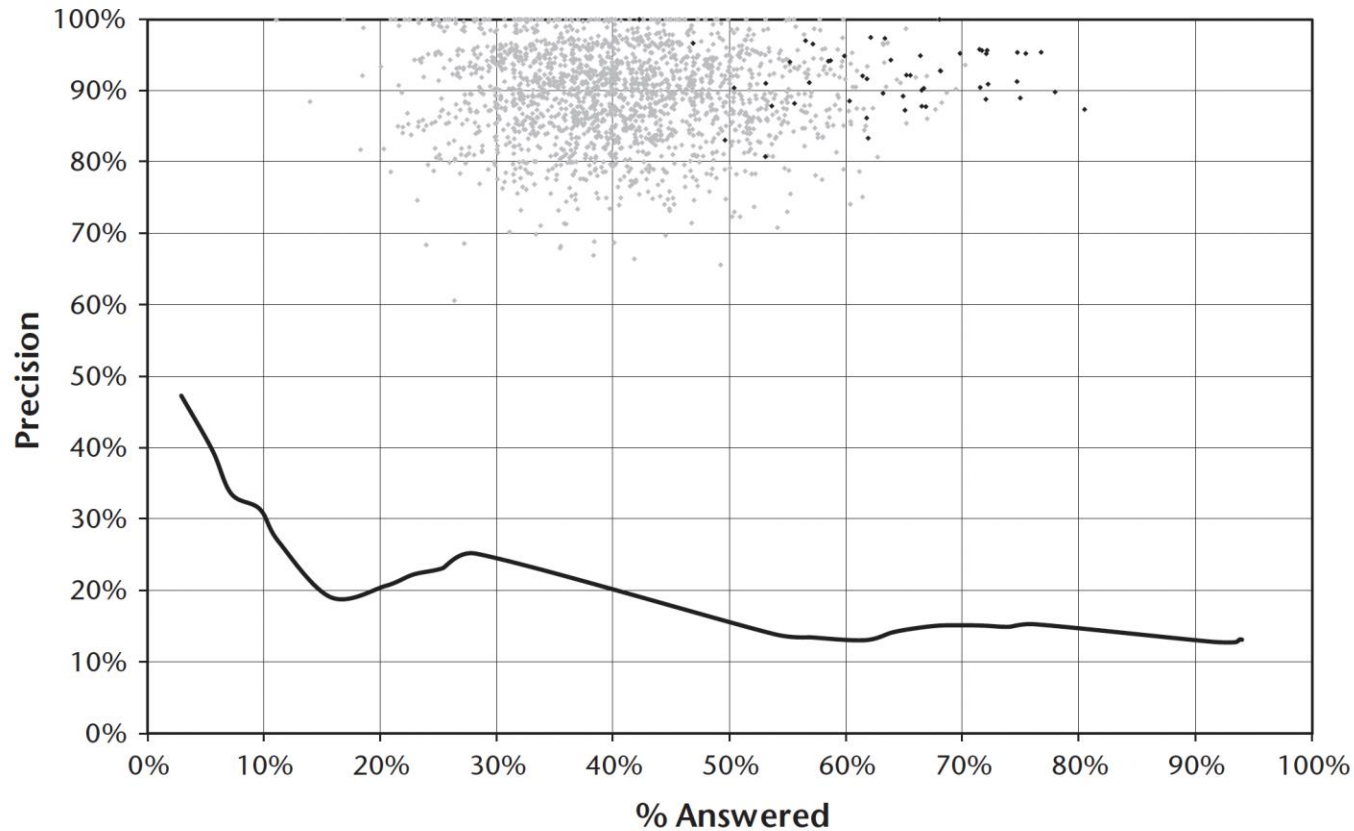


Figure 4. Baseline Performance.

The questions used were 500 randomly sampled Jeopardy clues from episodes in the past 15 years. The corpus that was used contained, but did not necessarily justify, answers to more than 90 percent of the questions. The result of the PIQUANT baseline experiment is illustrated in figure 4. As shown, on the 5 percent of the clues that PIQUANT was most confident in (left end of the curve), it delivered 47 percent precision, and over all the clues in the set (right end of the curve), its precision was 13 percent.

Benchmarks

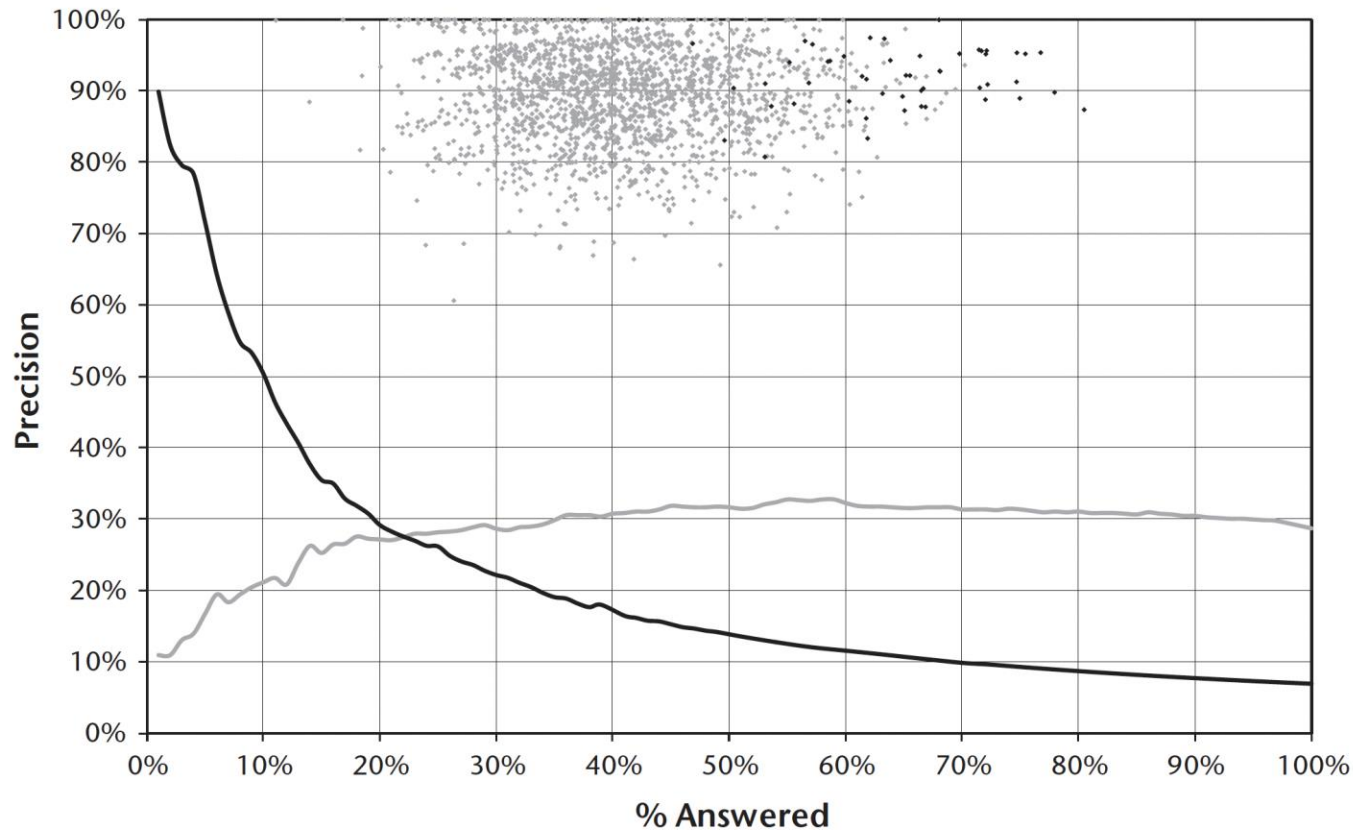


Figure 5. Text Search Versus Knowledge Base Search.

In figure 5 we show two other baselines that demonstrate the performance of two complementary approaches on this task. The light gray line shows the performance of a system based purely on text search, using terms in the question as queries and search engine scores as confidences for candidate answers generated from retrieved document titles. The black line shows the performance of a system based on structured data, which attempts to look the answer up in a database by simply finding the named entities in the database related to the named entities in the clue.

The Four Pillars of the DeepQA Architecture



Massive Parallelism

Exploring multiple interpretations simultaneously.



Many Experts

Integrating a wide range of loosely coupled probabilistic analytics.

	85%	85%	92%
85%	92%		45%
45%		92%	72%
	80%		60%

Pervasive Confidence

No single component commits to an answer; everything generates features and confidences.



Shallow & Deep Integration

Balancing strict semantics with shallow semantic ontologies.

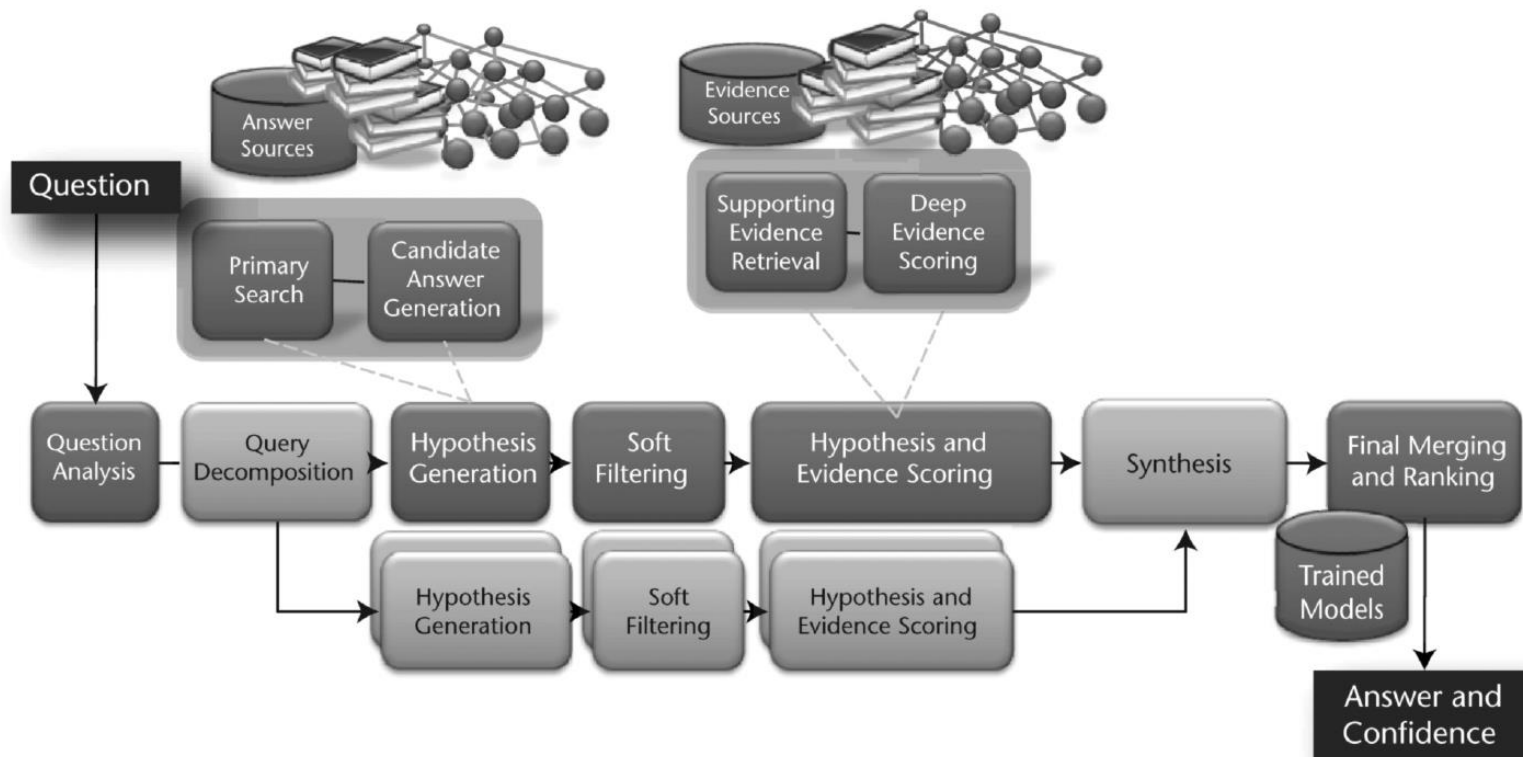
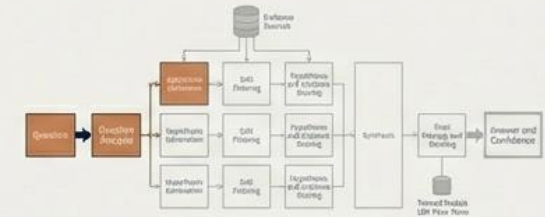
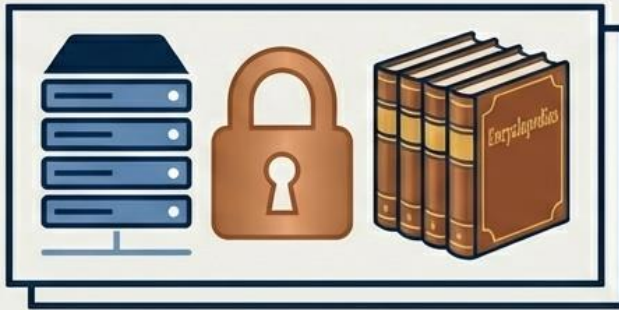


Figure 6. DeepQA High-Level Architecture.

Step 1 & 2: Content Acquisition & Question Analysis

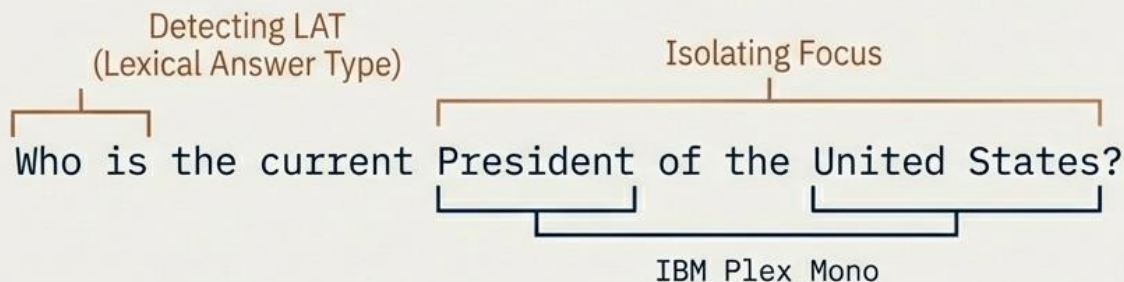


Acquisition

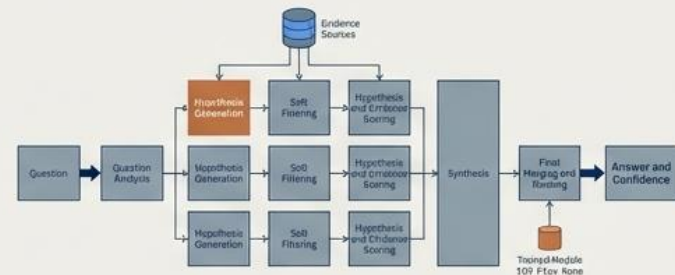
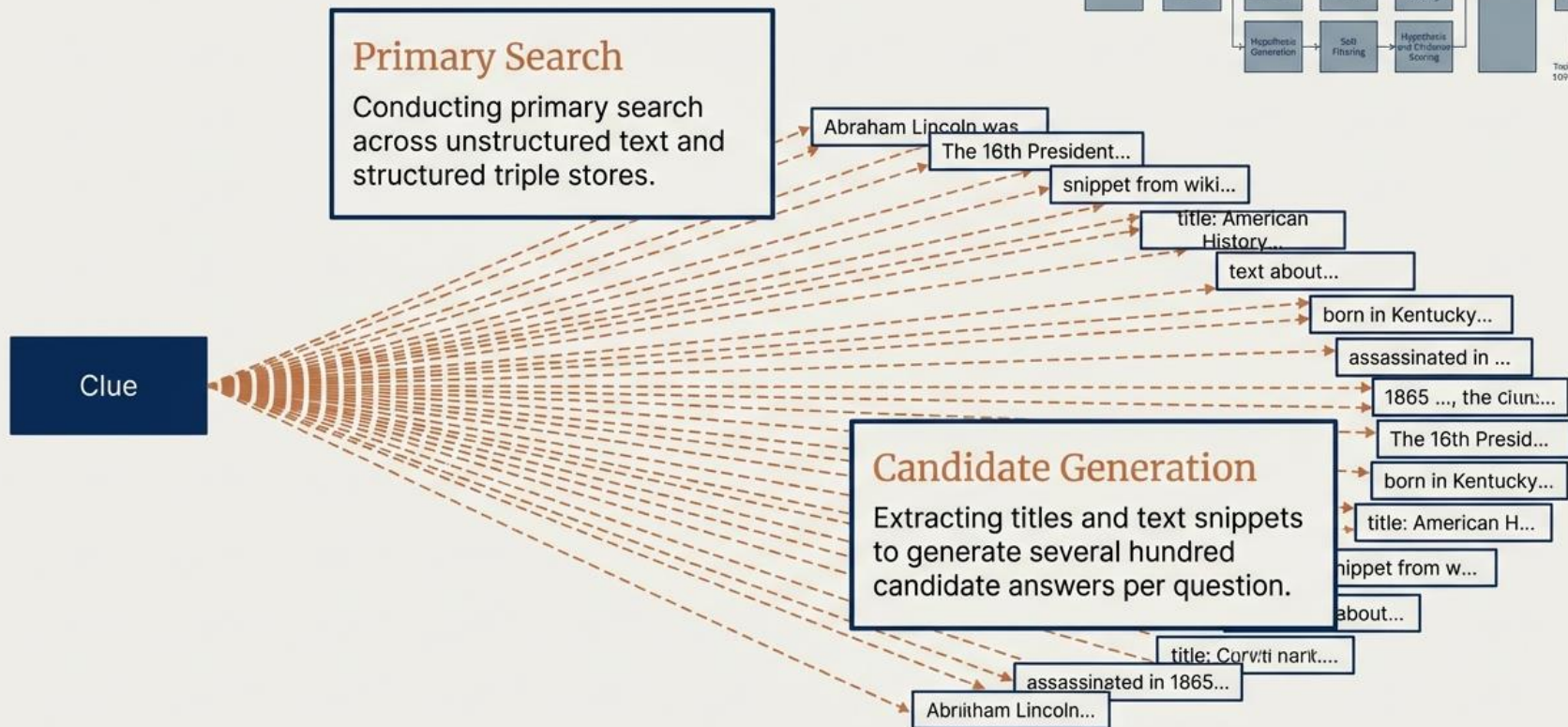
Building the corpus through encyclopedias, dictionaries, and newswire. Strict rule: System is completely self-contained during play. No live web access.

Analysis

Parsing the focus and Lexical Answer Type. Detecting semantic relations before generating a query.

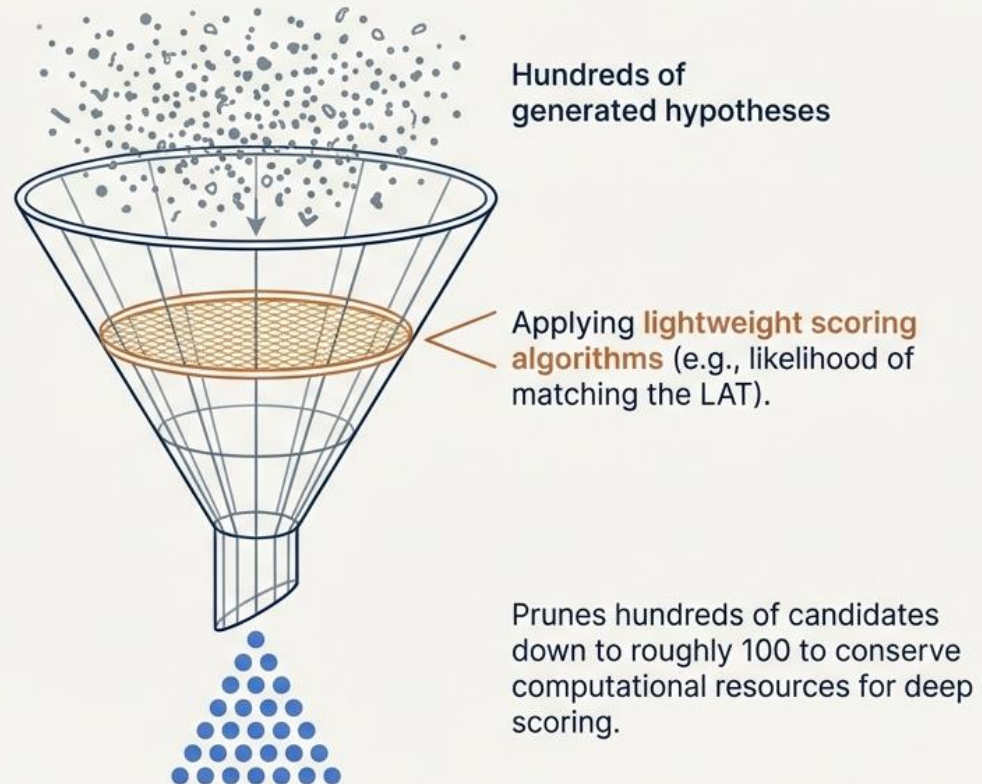
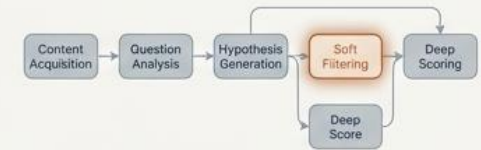


Step 3: Hypothesis Generation



Step 4: Soft Filtering the Noise

Minimap of the pipeline

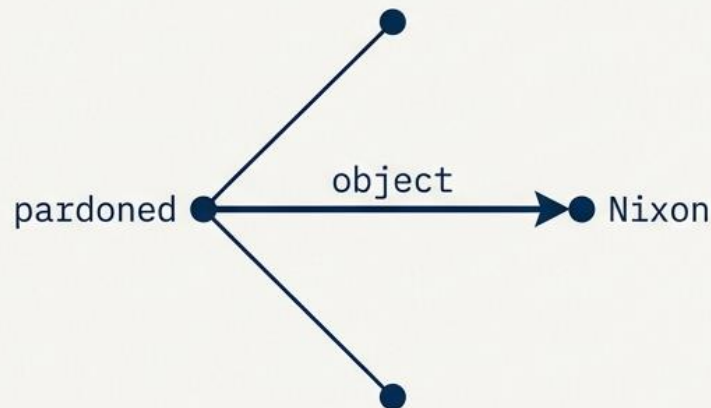


Step 5: Deep Hypothesis & Evidence Scoring



Passage Scoring

Clue: He was presidentially pardoned on Sept. 8, 1974.



Logical form alignment proves Nixon is the object of the pardon, beating simple keyword matches for Ford.

Evidence Profiles

Clue: Chile shares its longest land border with this country.

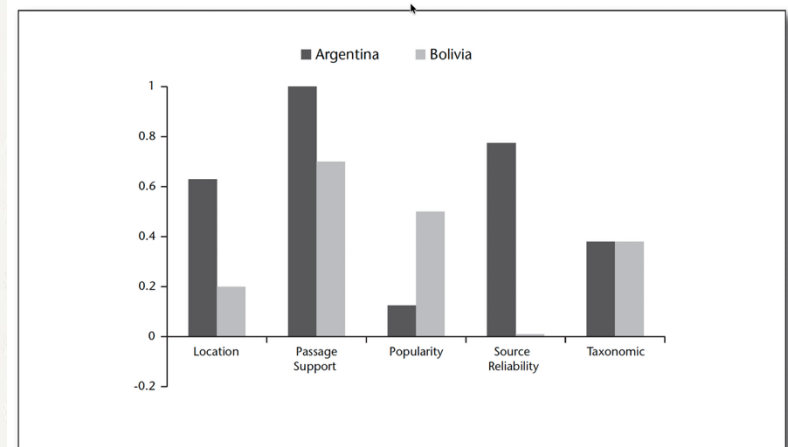
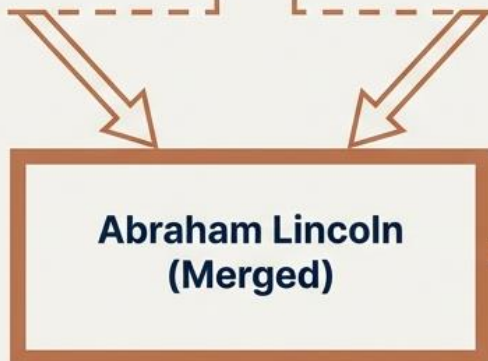


Figure 8. Evidence Profiles for Two Candidate Answers.

Argentina beats Bolivia not on popularity, but on Geospatial and Source Reliability dimensions.

Step 6: Final Merging & Ranking

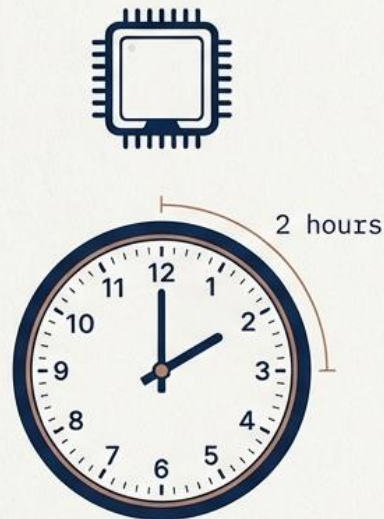


Combining disparately supported surface forms.



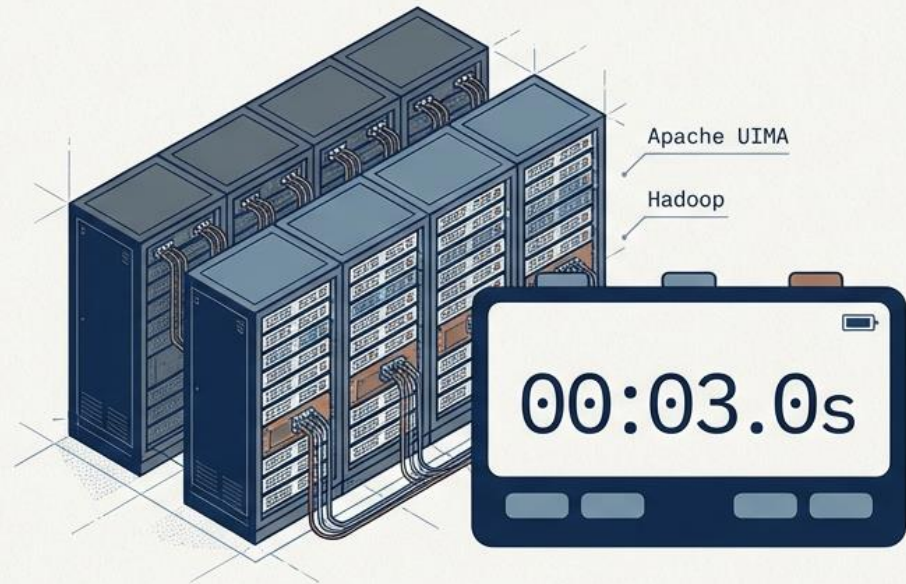
Analyzes thousands of feature scores to estimate the final likelihood of correctness and decide whether to buzz in.

The Engineering Feat: Scaling for Speed



1 Processor

2 hours to answer a single question.

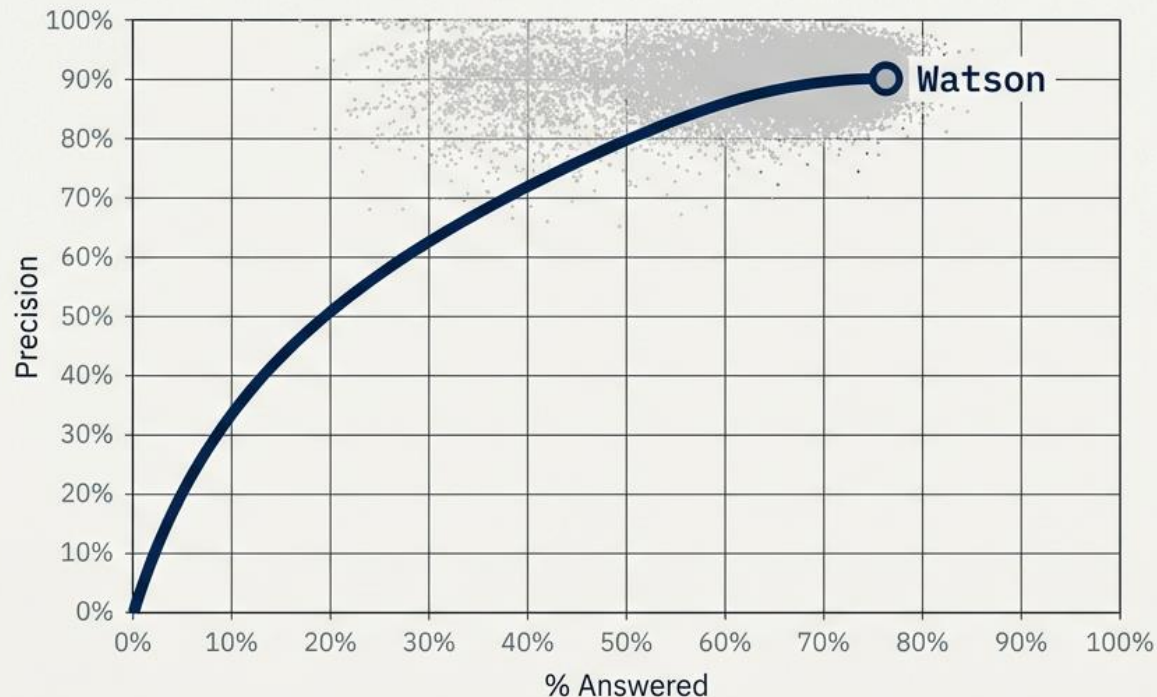


2,500 Compute Cores

Embarrassingly parallel workload driving latencies into the 3-5 second range.

Entering the Winners Cloud

Precision vs % Answered



- 3 years of intense R&D.
- 20 researchers and software engineers.
- A foundational architecture proving that machines can synthesize, evaluate, and answer complex human language at an expert level.

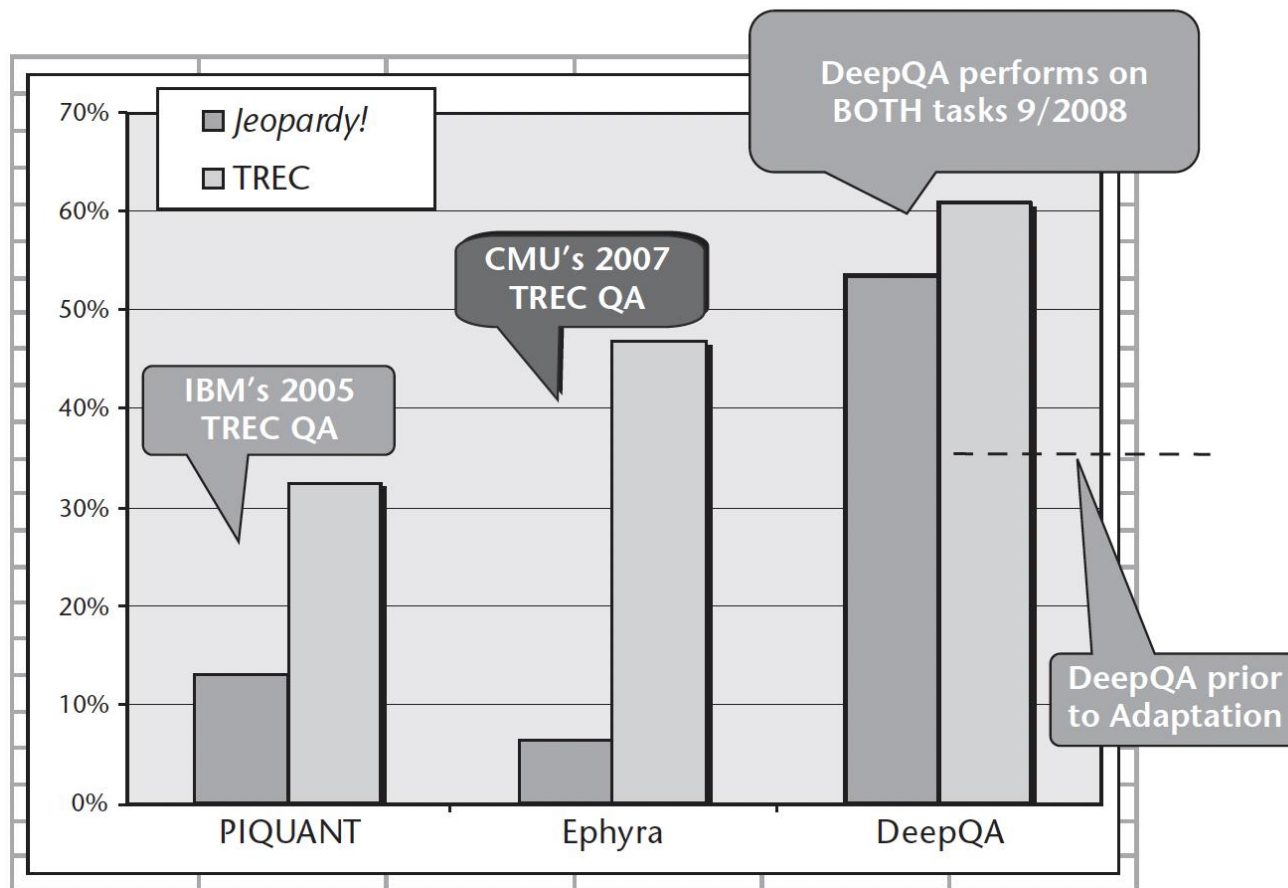


Figure 10. Accuracy on Jeopardy! and TREC.

Figure 10 shows the results of the adaptation experiment. Both the 2005 PIQUANT and 2007 OpenEphyra systems had less than 50 percent accuracy on the TREC questions and less than 15 percent accuracy on the Jeopardy clues. The DeepQA system at the time had accuracy above 50 percent on Jeopardy. Without adaptation DeepQA's accuracy on TREC questions was about 35 percent. After adaptation, DeepQA's accuracy on TREC exceeded 60 percent.

Project Outcomes and Performance

- The Jeopardy Challenge helped address requirements that led to the design of the DeepQA architecture and the implementation of Watson.
- After 3 years of intense research and development by a core team of about 20 researchers, Watson is performing at human expert levels in terms of precision, confidence, and speed at the Jeopardy quiz show.
- Results strongly suggest that DeepQA is an effective and extensible architecture that may be used as a foundation for combining, deploying, evaluating, and advancing a wide range of algorithmic techniques to rapidly advance the field of QA.

Systems-Level Approach

- No “one” algorithm solves challenging problems like this.
- The architecture and methodology developed highlighted the need to take a systems-level approach to research in QA, which applies to research in the broader field of AI.
- A system design and methodology that facilitated the efficient integration and ablation studies of many probabilistic components was essential.

Rapid Experimentation and Infrastructure

- Rapid experimentation was another critical ingredient to success.
- The team conducted more than 5500 independent experiments in 3 years, each averaging about 2000 CPU hours and generating more than 10 GB of error-analysis data.
- Without DeepQA's massively parallel architecture and a dedicated high-performance computing infrastructure, these experiments would not have been possible.

Current Status (by 2010)

- Tuned for the Jeopardy Challenge, Watson has begun to compete against former Jeopardy players in a series of "sparring" games.
- It is winning 64 percent of the games but has to be improved and sped up to compete favorably against the very best.

The Current Status of DeepQA (As of 2026)

- The original DeepQA architecture is now considered a legacy system. Used a rule-based, evidence-driven architecture & no deep neural networks
- Manual feature engineering and rule-based pipelines have been superseded by Large Language Models (LLMs) and Transformer-based deep learning networks.

Commercial Challenges: Watson Health (As of 2026)

- Attempts to commercialize the Jeopardy-optimized system for complex fields like oncology were unsuccessful.
- The system struggled to process unstructured, real-world clinical data, eventually leading IBM to sell off the Watson Health division.

Commercial Challenges: Watson Health (As of 2026)

- IBM spun off Watson Health into a separate company called Merative
- Sold to private equity firm Francisco Partners
- Marked the official acknowledgment that the Watson Health vision had failed

Watson → watsonx (As of 2026)

- The "Watson" brand continues, but the underlying 2010s QA technology does not.
- Today, IBM's flagship AI platform is rebranded under the name "watsonx", which is built on Generative AI, Foundation Models, and machine learning orchestration.
- New features include AgentOps: a full lifecycle tool for building and managing AI agents

Watson → watsonx (As of 2026)

- Collaborated with the Recording Academy on AI-powered experiences for the 2026 Grammys



- Also, IBM partnered with Anthropic to integrate Claude models into the watsonx platform...

How AI helps break the cost barrier to COBOL modernization

Legacy code modernization stalled for years because understanding legacy code cost more than rewriting it. AI flips that equation.

Category
[Claude Code](#)

Product
[Claude Code](#)

Date
February 23, 2026

Reading time
5 min

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MARKETS

IBM is the latest AI casualty. Shares tank 13% on Anthropic programming language threat

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